

Module 1 — Strings

Python for Beginners — ISDP Course

بِسْمِ اللَّهِ الرَّحْمَنِ الرَّحِيمِ

In the name of Allah, the Most Gracious, the Most Merciful

What You Will Learn in This Module

#	Concept	What it means
1	Strings	What text looks like in Python
2	Concatenation	Joining two strings together using +
3	input()	Asking the user to type something
4	len()	Counting how many letters are in a string

Duration: 2 classes (45–60 minutes each)

Prerequisites: Module 0 — you know what `print()` does

How to use this notebook

- Each section has an **explanation** followed by **runnable code cells**
 - Click on a code cell and press **Shift + Enter** to run it
 - Feel free to **change the code** and run it again — that is how you learn!
 - There are **Try It Yourself** cells throughout — fill them in and run them
-

Section 1 — What is a String?

A **string** is any piece of text inside quotation marks.

It can contain: - Letters: "Ahmed" - Words: "Assalamu Alaikum" - Sentences: "The Quran has 114 Surahs" - Numbers written as text: "786" - Urdu or Arabic: "السلام عليكم"

Python treats **everything inside quotation marks as text** — it will never try to calculate it.

Single quotes or double quotes?

Both work. "Ahmed" and 'Ahmed' are exactly the same.

Pick one style and stay consistent.

We will learn about numbers vs strings in **Module 3 (Data Types)**.

```
# Run this cell – these are all strings
print("Assalamu Alaikum")
print("Pakistan")
print("123")           # Even numbers become text when inside quotes
print("!")
print("Surah Al-Fatiha has 7 verses")

# Single quotes and double quotes both work
print("This uses double quotes")
print('This uses single quotes')
# Both lines print the same kind of output

# Strings can be short or very Long
print("Hi")
print("The Holy Quran was revealed to Prophet Muhammad (PBUH) over 23 years")
print("Lahore")
print("May Allah bless you and your family. Ameen.")

print()
print("You now understand what a string is. Ma sha Allah!")
```

Try It Yourself — Task 1

Print your own name.

```
# Task 1
```

Try It Yourself — Task 2

Print the name of your city.

```
# Task 2
```

Try It Yourself — Task 3

Print this sentence exactly:

```
# Task 3
```

Try It Yourself — Task 4

Print a greeting in Urdu or Arabic.

```
# Task 4
```

Try It Yourself — Task 5

Print a sentence about your favourite Prophet (peace be upon them).

```
# Task 5
```

Section 2 — String Concatenation (Joining Strings)

Concatenation means joining two or more strings into one. We use the + symbol to join strings.

Think of it like gluing words together side by side.

```
"Assalamu" + "Alaikum" → "AssalamuAlaikum"
```

⚠ Spaces must be added manually

The + operator joins **exactly** what you give it — nothing more.

Code	Output
"Assalamu" + "Alaikum"	AssalamuAlaikum ❌
"Assalamu " + "Alaikum"	Assalamu Alaikum ✅
"Assalamu" + " Alaikum"	Assalamu Alaikum ✅

The space belongs inside one of the strings.

```
# Basic concatenation
print("Assalamu" + "Alaikum")
print("Maulana " + "Abdul Rahman")
print("Lahore" + ", " + "Pakistan")

# See what happens when you forget the space
print("Assalamu" + "Alaikum")      # Wrong – no space
print("Assalamu " + "Alaikum")     # Correct – space at the end of first string
print("Assalamu" + " Alaikum")     # Also correct – space at the start of second string

# Practical examples with Pakistani/Islamic context
print("Welcome, " + "Ustaz Ahmad!")
print("In the name of " + "Allah, the Most Gracious, the Most Merciful")
print("Today's lesson: " + "Introduction to Python")
print("I am from " + "Karachi" + ", Pakistan")
print("The Quran has " + "114" + " Surahs and " + "30" + " Paras")
```

Concatenation with Urdu and Arabic

Python supports all languages including **Urdu** and **Arabic**. The same + operator works exactly the same way.

Note: In the terminal, right-to-left text may look slightly different, but the logic is identical to English strings.

```

# Urdu strings
print("السلام " + "عليكم")
print("خوش آمدید " + "میرے پیارے طالبعلموں")
print("مولانا " + "عبدالرحمن")
print("سیکھ رہے ہیں " + "Python ")
print("محنت " + "کامیابی کی کنجی ہے")
print("علم حاصل کرنا " + "ہر مسلمان پر فرض ہے۔")

# Arabic strings
print("بِسْمِ اللّٰهِ " + "الرَّحْمٰنِ الرَّحِیْمِ")
print("اقْرَأْ بِاسْمِ رَبِّكَ " + "الَّذِي خَلَقَ")

# Mixing English and Urdu
print("سیکھ رہے ہیں " + "Python ")
print("My name is " + "احمد علی")
print("Subject: " + "علوم القرآن")

# Urdu + English
# English + Urdu
# English Label + Urdu value

print()
print("Excellent! Python speaks Urdu and Arabic too. Alhamdulillah!")

```

⚠ Common Mistake — You cannot + a string and a number

This is one of the most common errors beginners make:

```
print("There are " + 5 + " prayers") # ✗ TypeError!
```

Python does not know how to glue the **text** "There are " to the **number** 5.
You must first convert the number to a string using **str()**.

Two ways to fix this:

Method	Code	Notes
str() conversion	"There are " + str(5) + " prayers"	Works everywhere
Comma in print()	print("There are", 5, "prayers")	Easier but adds spaces automatically

This will cause a TypeError – uncomment the next line to see the error
print("There are " + 5 + " prayers")

Fix 1 – use str() to convert the number to a string
print("There are " + str(5) + " prayers")

Fix 2 – use a comma inside print() instead of +
print("There are", 5, "prayers")

Another example
print("Pakistan was created in " + str(1947))
print("The Quran has " + str(114) + " Surahs")

Try It Yourself — Task 1

Use to build and print this sentence (join at least 3 parts):

Task 1

Try It Yourself — Task 2

Print this sentence using for the number:

Task 2

Try It Yourself — Task 3

Build and print a greeting in Urdu using two string parts joined with .

Task 3

Using Quotes Inside a String

What if the text itself contains a quotation mark?

Trick 1 — Use the other type of quote

If your text contains ' , wrap the whole string in " (double quotes). If your text contains " , wrap the whole string in ' (single quotes).

You want to print	Use this
Ahmed's book	"Ahmed's book"
She said "Bismillah"	'She said "Bismillah"'

Trick 2 — Escape character \

You can also put a backslash \ before the quote to tell Python “this is part of the text, not the end of the string”:

```
print('Ahmed\'s book')
print("She said \"Bismillah\"")
```

```
# Using the other type of quote
print("Ahmed's book")
print('She said "Bismillah"')
```

```
# Using the backslash escape character
print('Ahmed\'s book')
print("She said \"Bismillah\"")
```

Special Characters — \n and \t

Some characters cannot be typed directly into a string — instead we use a **backslash code**:

Code	Name	Effect
\n	newline	Moves to the next line
\t	tab	Adds a tab space (like pressing the Tab key)

```
print("Lahore\nKarachi")  
# Lahore  
# Karachi  
  
print("Name:\tAhmed")  
# Name:      Ahmed
```

Even though \n and \t look like two characters, Python treats each as **one** special character.

```
# \n moves to a new line  
print("Lahore\nKarachi\nIslamabad")  
  
# \t adds a tab space – useful for lining up text  
print("Name:\tAhmed")  
print("City:\tLahore")  
  
# Using both together  
print("Surah:\tAl-Fatiha\nVerses:\t7")
```

Try It Yourself — Task A

Print this sentence exactly, including the apostrophe:

Hassan's school is in Multan.

```
# Task A
```

Try It Yourself — Task B

Using **one** print() statement and \n, print these three lines:

```
Quran  
Hadith  
Fiqh
```

Then, using **one** print() statement and \t, print:

```
Subject:[TAB]Arabic
```

```
# Task B
```

Drawing Patterns with `print()`

We can use symbols like `*`, spaces, and `\n` to draw simple pictures with text. This is called **ASCII art** — pictures made out of characters.

```
print(" *")
print("* *")
print("***")
```

Output:

```
*
* *
***
```

Each `print()` is one row. Spaces at the start push the symbols to the right.

```
# A simple star triangle
```

```
print(" *")
print(" * *")
print("*****")
```

```
print()
```

```
# A small box using * and spaces
```

```
print("*****")
print("*      *")
print("*****")
```

Try It Yourself — Task C

Draw the **first letter of your name** using stars (`*`) and spaces.

For example, the letter **A**:

```
*
* *
*****
*      *
*      *
```

Use one `print()` for each row. Count your spaces carefully!

```
# Task C
```

Section 3 — `input()` — Getting Information from the User

So far, all the information in our programs has been **fixed** — we wrote it ourselves. With `input()`, we can ask the **user** to type something while the program is running.

How input() works

```
print("Assalamu Alaikum, " + input("Please enter your name: "))
```

1. Python **stops** the program
2. It **shows** the text inside input("...") as a prompt
3. It **waits** for the user to type and press Enter
4. Whatever the user typed is used **right there**, as part of the print()

⚠ Important rule — input() always returns a STRING

Even if the user types 25, Python receives it as "25" (text, not a number). We will learn how to convert it to a real number in **Module 3**.

```
# Basic input – ask name and greet
```

```
print("Assalamu Alaikum, " + input("Please enter your name: "))
```

```
# Ask for city
```

```
print("Ma sha Allah, " + input("Which city are you from? ") + " is a wonderful city!")
```

```
# Ask for favourite subject
```

```
print("Great choice! " + input("What is your favorite subject? ") + " is very important.")
```

input() works directly inside print()

As you saw above, input("...") can be placed **directly inside** print(...), joined with + just like any other string. Python asks the question, waits for the answer, and immediately uses it.

```
# input() directly inside print()
```

```
print("Wa Alaikum Assalam, " + input("Who are you replying to? "))
```

```
# Islamic and Pakistani context examples
```

```
print("May Allah give barakah to " + input("What is the name of your school or madrasa? ") + "!!")
```

```
# Favourite Prophet
```

```
print("Peace and blessings be upon Prophet " + input("Name your favorite Prophet (peace be upon them): "))
```

```
# input() always returns a STRING – even when the user types a number
```

```
print("You are " + input("How old are you? ") + " years old. May Allah bless you!")
```

```
# Notice: we can join the answer with strings using + because input() gives back a string
```

Asking multiple questions

You can call input() as many times as you need — each call asks one question.

A small preview: If we want to **combine** the answers to two different questions into one sentence, we need a way to give each answer a name so we can use it again later. That “naming” idea is called a **variable** — the full lesson is in **Module 2**. For now, just notice how useful it would be to remember an answer!

```
# Ask first name and last name, print them joined
first = input("Enter your first name: ")
last = input("Enter your last name: ")
print("Assalamu Alaikum, " + first + " " + last + "!")

# A simple introduction – ask questions one by one
name = input("Your name: ")
print("My name is " + name + ".")

city = input("Your city: ")
print("I am from " + city + ".")

school = input("Your school or madrasa: ")
print("I study at " + school + ".")
```

Try It Yourself

Build a **Madrasa Registration** program that: 1. Asks for the student’s name, age, class level (e.g. Hifz / Nazra / Alim), and favourite Islamic subject 2. Prints a formatted registration card like this:

```
=====
MADRASA REGISTRATION
=====
Name           : Ahmed Ali
Age            : 14
Class Level    : Hifz
Fav Subject    : Tafseer
=====
Registration complete. Jazakallah Khair!
```

Write your Madrasa Registration program here

Section 4 — len() — Counting Letters in a String

len() stands for *length*. It counts how many **characters** are in a string.

A character is any single letter, number, space, or symbol.

```
len("Ahmed")      → 5
len("Alhamdulillah") → 13
len("Hi")         → 2
len("A B")        → 3 (the space counts too!)
```

How to use len()

```
print(len("Ahmed"))
```

Remember: len() gives back a **number** (integer). To join it with a string using +, you must wrap it in str(): str(len("Ahmed"))

```
# Basic len() examples
```

```
print(len("Ahmed"))      # 5
print(len("Alhamdulillah")) # 13
print(len("Pakistan"))   # 8
print(len("Quran"))       # 5
```

```
# Spaces count as characters too
```

```
print(len("Ali"))        # 3
print(len("Ali Khan"))    # 8 (space counts!)
```

```
# len() with a string directly
```

```
print("The word Bismillah has " + str(len("Bismillah")) + " letters")
print("The word Alhamdulillah has " + str(len("Alhamdulillah")) + " letters")
```

```
# len() with input()
```

```
print("Your name has " + str(len(input("Enter your name: "))) + " letters.")
```

Another preview: The next example uses the same name twice (once to greet, once to count letters). To reuse an answer, we give it a name — a **variable**. We'll fully learn this in **Module 2**.

```
# Practical example – check name length
```

```
name = input("Enter your full name: ")
print("Assalamu Alaikum, " + name + "!")
print("Your full name has " + str(len(name)) + " characters (including the space).")
```

Try It Yourself — Task 1

Print the length of the word Bismillah. Count the letters yourself first, then check with len().

```
# Task 1
```

Try It Yourself — Task 2

Ask the user for their city name. Print: "Your city name has [n] letters."

```
# Task 2
```

Try It Yourself — Task 3

Ask for first name and last name separately (two input() calls). Print the length of each one.

Section 5 — Common Mistakes Summary

These are the most frequent mistakes in this module. Study each one carefully.

#	Mistake	Wrong	Fixed
1	Forgetting space between words	"Ahmed"+"Khan" → AhmedKhan	"Ahmed " + "Khan"
2	Joining string + number with +	"Age: " + 15 → TypeError	"Age: " + str(15)
3	Missing quotes in input()	input(Enter name:) → SyntaxError	input("Enter name:")
4	Forgetting input() returns a string	"Age: " + input("Age: ") → works fine, but it's text not a number	Remember to convert with int() later (Module 3)
5	Using len() result directly with +	"Length: " + len(x) → TypeError	"Length: " + str(len(x))

Mistake 1 – forgetting space

Wrong:

```
print("Ahmed" + "Khan")           # AhmedKhan
```

Fixed:

```
print("Ahmed " + "Khan")          # Ahmed Khan
```

Mistake 2 – joining string and number

Wrong (uncomment to see the error):

```
# print("Age: " + 15)
```

Fixed:

```
print("Age: " + str(15))
```

Mistake 3 – missing quotes in input()

Wrong (uncomment to see the error):

```
# print(input(Enter your name: ))
```

Fixed:

```
print(input("Enter your name: "))
```

Mistake 5 – using len() result with +

Wrong (uncomment to see the error):

```
# print("Length: " + len("Bismillah"))
```

Fixed:

```
print("Length: " + str(len("Bismillah")))
```

Section 6 — Module Summary

What you have learned

Concept	Syntax	Example
Print a string	<code>print("text")</code>	<code>print("Assalamu Alaikum")</code>
Join strings	<code>"a" + "b"</code>	<code>"Lahore" + ", Pakistan"</code>
Number to string	<code>str(n)</code>	<code>str(114) → "114"</code>
Get user input	<code>input("prompt")</code>	<code>input("Your name: ")</code>
Count characters	<code>len(x)</code>	<code>len("Ahmed") → 5</code>

Key Rules to Remember

1. Text in quotes = string. "786" is text, 786 is a number.
 2. + joins strings — spaces must be added manually.
 3. You cannot use + between a string and a number — use `str()` first.
 4. `input()` **always** gives back a string — even if the user types a number.
 5. `len()` returns a number — use `str(len(...))` when joining with text.
-

Final Practice — Put It All Together

Build a complete **“About Me” program** that: 1. Asks for name, city, school/madrassa, age, and favourite Islamic subject 2. Prints a neat, formatted introduction using all the answers 3. Also prints: "Your name has [n] letters."

Note: Since this program needs to use the same answer (like your name) more than once, you will need to give each answer a name — a **variable**. This is a small preview of Module 2: `name = input("Your name: ")`

Example output:

```
Bismillah!
-----
My name is Ahmed Ali.
I am from Lahore.
I study at Jamia Islamia.
I am 14 years old.
My favourite subject is Tafseer.
-----
My name has 9 letters.
JazakAllah Khair!
```

Your final program here

Assessment Checklist

Before moving to Module 2, make sure you can do all of these **without looking at the notes**:

- ☐ Write a string and print it
- ☐ Join two strings using + with correct spacing
- ☐ Use `str()` to join a number into a string
- ☐ Use `input()` to ask the user a question and store the answer
- ☐ Use `len()` to count letters in a string
- ☐ Build a small program that asks 2–3 questions and prints a summary

Oral Questions (your teacher may ask these)

1. How do you join two strings in Python? *(Answer: using +)*
2. What does `input()` do? *(Answer: pauses the program and waits for the user to type)*
3. What does `len("Ahmed")` return? *(Answer: 5)*
4. Why does `"Age: " + 15` give an error? *(Answer: you cannot join a string and a number with + directly)*

الحمد لله — You have completed Module 1!

Next up: **Module 2 — Variables** 🎉